

Vertical Integration Means Silicon Opportunity

Silicon platforms help to enable vertical integration by bringing high-performance digital, analogue, memory and radio frequency (RF) technologies—all of which have historically been separate on a single chip

PHIL BISHOP
CORPORATE VICE PRESIDENT OF WORLDWIDE MARKETING
MAGMA DESIGN AUTOMATION

During start-up period, a company has to do everything itself, from providing all the pieces of the technology (hardware and software) to bundling it together into a solution. This is vertical integration, and it allows the company to control the customer's experience, leaving nothing to chance in those early make-or-break days.

In electronics, when an entire technology product family is managed vertically, it must be based on a solid silicon platform—a system-on-chip (SoC). It is on this chip that all the necessary digital, analogue and memory resources reside to support the specific applications that will be running. There are three markets in particular (refer Fig. 1) that drive platform development: Mobile technology, cloud computing and consumer electronics. Each of these has different needs.

Mobile products are proliferating wildly, and their utility is extending far beyond simple communication. In particular, streaming video is becoming commonplace, and this puts enormous stress on the phone's graphics capabilities. Mobile gaming and higher user expectations for resolution and clarity intensify the pressure on the graphics subsystem; resolution is now at least 720p, and often as high as 1080p.

Connectivity is the other critical resource for a mobile device. Smartphones must have WiFi and 3G/4G cellular capabilities. Such devices can

also typically connect to servers and PCs in order to access more permanent information, enable file sharing and download data. This means that mobile system-on-chip (SoC) platforms need to bring together multi-core CPU, graphics processor, USB, radio modem and multimedia processing on a single chip. Performance and power are, of course, critical; symmetric multiprocessing helps to manage the balance.

A cloud computing environment, in contrast, is one where computing resources can be pooled and accessed remotely on demand, and it has different requirements. This means that the communication of data into and out of such an environment can define the user experience. Dedicated network processing SoC platforms

offload the burden of ensuring that the network traffic (wired, wireless or mobile) is handled, while providing enough high-speed memory to store all the transient data and managing security. This allows the standard server subsystem to handle the actual applications doing the work that the user requested.

Finally, consumer electronics is characterised in part by its sheer variety—mobile devices, high-end gaming, high-definition TV, digital cameras, media players and advanced home appliances. Here again, performance and connectivity are critical, and emerging SoC platforms have to be able to manage high-resolution video balanced against connectivity, performance, energy efficiency and cost.

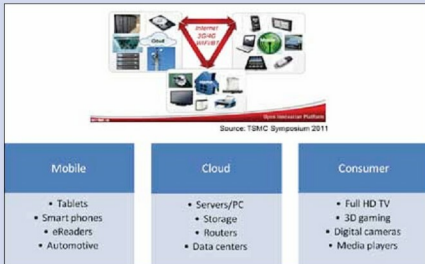


Fig. 1: Emerging silicon market opportunities